PRECISION DRIVEN FARMING

Problem Definition

Small-scale and commercial farmers have problems identifying crop diseases early and using pesticides in the right way. Crop diseases can spread quickly if they are not noticed early. This can damage crops, reduce the amount of food produced and cause farmers to lose money. Normally, farmers inspect their fields manually, which can take a lot of time, especially on large farms. It can also be difficult to check every plant properly. Farmers sometimes spray pesticides over the whole field even when only a small part of the field is affected. This can waste pesticides, increase farming costs and cause chemicals to enter the soil and nearby water sources. Because of this, farmers need a faster and more effective way to monitor crops and identify diseases early.

AI Benefit

Precision Driven Farming can help by using cameras on drones or fixed poles to take pictures of crops and leaves. Computer vision can then be used to analyse these images and identify possible crop diseases. The system can also create a spraying map that shows which areas may need treatment. This can help farmers use less pesticide, reduce costs and protect their crops. For the municipality and local community, healthier crops can help increase local food production and support farmers. Using less pesticide can also help reduce chemical runoff into nearby water sources and support a cleaner environment.

BUSINESS BACKGROUND:

Industry Background

Agriculture contributes 2.7% to SA GDP and employs 860,000 people, but faces critical challenges. Over 30% of crop yield is lost annually to pests and diseases, with late detection being the #1 cause. Traditional farming relies on manual scouting where a worker walks the field and visually checks leaves. This method is slow, covers only 5-10% of a large field, and misses early-stage infections which are invisible to the naked eye.

At the same time, chemical costs have increased by 45% since 2022.

Farmers practice blanket spraying - spraying the entire field even if only 20% is infected - leading to high costs, chemical resistance, and environmental damage.

1. **Company / Project Background**
    - Precision Driven Farming is conceptualized as an AgriTech start-up focused on small to medium commercial farms in the country. The business was founded to solve a simple problem observed on local farms
    - Farmers know they have a disease only when 40% of the crop is already infected.
    - The project leverages the 4th Agricultural Revolution (Agriculture 4.0) which combines AI, drones, and IoT.
    - Unlike expensive solutions from John Deere or Trimble that cost >R400k and target US mega-farms
    - It is built for the African context: low-cost, works with a smartphone, and uses WhatsApp for alerts which every farmer already uses.

2. **Business Context & Problem Drivers**

- Economic Driver: Fertilizer & pesticide costs are farmer's largest expense at 35% of input costs.
- Environmental Driver: South Africa's new Agricultural Pest Act pressures farmers to reduce chemical runoff
- . Technological Driver: 92% of SA farmers now have smartphones and cheap drones (<R5k) are available.
- Social Driver: Shortage of skilled agronomists - one agronomist serves 100+ farms, cannot visit often.

3. Current Business Process (AS-IS)

- Farmer notices yellowing/brown spots
- Takes photo and sends to agronomist on WhatsApp
- Waits 2-3 days for response
- Agronomist visits, confirms Late Blight
- Farmer sprays entire 100ha field
- Result: Disease has spread, chemical wasted

4. Proposed Business Process (TO-BE)

- Drone / fixed poles cameras captures 100 images per 10ha daily
- AI automatically detects Early Blight at 5% severity
- System generates Spray Map: Zone A=0L/ha, Zone B=50L/ha, Zone C=150L/ha (Critical
- Farmer receives WhatsApp alert with map: "Spray Zone C within 24h"
- Tractor with variable-rate sprayer executes map OR manual spot-spraying
- Result: Infection contained at 5%, 65% chemical save

Business Objectives

The main business objective of Precision Driven Farming is to help small to medium commercial farmers in South Africa detect crop diseases early and treat only the affected areas, so that crop losses, chemical costs and environmental damage are reduced. The specific objectives are:

1. **Detect disease early.** Identify crop disease at an early stage (about 5% severity) instead of the current situation where farmers only discover disease once about 40% of the crop is infected.
2. **Reduce chemical use and cost.** Replace blanket spraying with a zone-based spray map, so that pesticide is applied only where it is needed.
3. **Speed up decision-making.** Deliver a disease alert and spray map to the farmer within 24 hours of image capture, instead of the 2 to 3 days currently spent waiting for an agronomist's response.
4. **Make the solution affordable and accessible.** Provide a low-cost system that works with a smartphone, low-cost drones and WhatsApp alerts, in contrast to existing solutions that cost over R400,000.
5. **Support, not replace, the farmer.** Keep the farmer in control by showing confidence scores and requiring the farmer to verify results before spraying.
6. **Benefit the community and environment.** Protect local food production and reduce chemical runoff into soil and nearby water sources.

Business Success Criteria

The project will be considered successful if the following measurable targets are met when the prototype is evaluated:

#	Criterion	Target	How it will be measured
1	Disease classification accuracy	At least 90% accuracy on unseen test images	Held-out test set, confusion matrix
2	Missed detections	Recall of at least 90% for diseased leaves	Test set recall per disease class
3	Early detection	Disease detected at an early stage severity	Test images labelled by severity
4	Processing time	Prediction within 10 seconds per image	Timed runs of the model
5	Alert speed	Alert and spray map delivered within 24 hours of capture	End-to-end test of the workflow
6	Chemical reduction	At least 50% less chemical than blanket spraying	Comparison of spray-map area against whole-field area on sample fields
7	Affordability	Hardware cost per farm under R10,000	Cost breakdown of drone and equipment

#	Criterion	Target	How it will be measured
8	Usability	At least 80% of test users can read the result and spray map without help	Short user test
9	Farmer verification	Every spray recommendation can be reviewed before action	Demonstration of the review step
10	Weather forecast	forecast MAE at least [20]% lower than the seasonal naive baseline	measured on the held-out final weeks

Functional Requirements

The system should be able to:

1. Take pictures of crops using drones or fixed cameras.
2. Process the images using Python and OpenCV.
3. Find and analyse leaves in the images.
4. Identify crop diseases using a PyTorch AI model.
5. Identify the affected areas of the crop.
6. Give disease predictions soon after an image is taken.
7. Record the time and location where a disease is detected.
8. Use GPS information to connect disease detections to areas in the field.
9. Create a map showing areas where pesticide treatment may be needed.
10. Show the results to farmers through a simple interface.
11. Save previous results so farmers can compare crop health over time.
12. Give a confidence score or warning when a disease is detected.

13. Allow the farmer to check the results before making a spraying decision.
14. Forecast likely outbreak weeks

Non-Functional Requirements

Requirement	Description
Accuracy	The system should identify crop diseases correctly as much as possible.
Performance	The system should process images quickly enough for near-real-time monitoring.
Reliability	The system should work properly during normal crop monitoring.
Scalability	The system should be able to work on both small and larger farms.
Usability	The system should be simple enough for farmers to understand and use.
Security	Farm information, images and user details should be protected from unauthorised access.
Maintainability	The software and AI model should be easy to update and fix when needed
Compatibility	The system should work with the chosen cameras, drones, Python, OpenCV and PyTorch.
Availability	The system should be available when crop monitoring is being carried out.
Explainability	The system should show the detected disease and, where possible, how confident the AI is about the result.

Constraints

1. Limited training data: It may be difficult to find enough good images of different crop diseases to train the AI.

2. **Image quality:** Poor lighting, blurry pictures, shadows, wind and the position of leaves can affect the results.
3. **Hardware limitations:** Drones and cameras have limits such as battery life, storage and processing power.

4. **Internet connection:** Some farming areas may have weak or no internet connection.

5. **Computing power:** Training the AI model may require a powerful computer or GPU.

6. **Weather:** Rain, strong wind, fog and dust can make it difficult to collect good images.

7. **Different crops:** Different crop types and growth stages may make disease detection more difficult.

8. **Cost:** Drones, cameras, maintenance and computer equipment can be expensive.

9. **Legal requirements:** Drone use and pesticide spraying must follow the relevant laws and regulations.

10. **AI limitations:** The AI can make mistakes, especially when it sees images or diseases that were not included in its training data.

11. no public time-stamped disease data exists so the disease log will be simulated

Risks and Mitigation Risk	Possible Impact	How to Reduce the Risk
Incorrect disease classification	The farmer could use the wrong treatment.	Train the AI using a large range of good-quality images and allow the farmer to check the result.
False disease detection	Pesticide may be used when it is not needed.	Use confidence levels and check uncertain results before spraying.
Missed disease detection	A disease could spread to other crops.	Monitor crops regularly and improve the AI model over time.
Poor image quality	The AI may give incorrect results.	Use suitable cameras and check the quality of images.
Drone failure	Some areas may not be monitored.	Maintain the drone and have another monitoring method available.
Battery runs out	The drone may not finish checking the field.	Plan the flight properly and have charged spare batteries.
Biased training data	The AI may not work well on some crops or environments.	Use different crops, diseases, environments and lighting conditions when training the model.
Data security breach	Farm information or images could be accessed by other people.	Use passwords, access controls and secure data storage.
Bad weather	Crop images may not be collected properly.	Monitor weather conditions and only fly

		when conditions are suitable.
Over-reliance on AI	Farmers could make a bad decision based only on the AI result.	Use the AI as a support tool and allow human checking before important decisions.
Incorrect spraying map	Pesticide could be sprayed in the wrong location.	Check the GPS information and allow the farmer to verify the map.
Regulatory problems	The project may break drone or pesticide regulations.	Make sure the system follows the relevant laws and regulations.

Summary

This project uses AI to solve a real problem in the farming industry. The system uses computer vision and deep learning to help farmers find crop diseases and identify areas that may need treatment. It does not replace the farmer but helps the farmer make better decisions. By using pesticides more carefully, farmers can reduce waste and costs while protecting their crops. The project can also benefit the local community by supporting food production, reducing chemical pollution and helping the local farming economy.

REFERENCES

Industry background - https://www.sanews.gov.za/south-africa/sa-agriculture-posts-record-exports-despite-global-headwinds